

```
In [2]: import os
import flexynesis
import torch
import numpy as np
import seaborn as sns
import pandas as pd
import random
import lightning as pl
```

Seed set to 42

1. Download and unpack the data:

https://bimsbstatic.mdc-berlin.de/akalin/buyar/flexynesis-benchmark-datasets/ccle_vs_gdsc.tgz

```
In [3]: if not os.path.exists("ccle_vs_gdsc"):
!wget -O ccle_vs_gdsc.tgz "https://bimsbstatic.mdc-berlin.de/akalin/b
--2025-03-14 14:12:53-- https://bimsbstatic.mdc-berlin.de/akalin/buyar/fl
exynesis-benchmark-datasets/ccle_vs_gdsc.tgz
141.80.181.46, 141.80.181.47rlin.de (bimsbstatic.mdc-berlin.de)...
connected. to bimsbstatic.mdc-berlin.de (bimsbstatic.mdc-berlin.de)|141.8
0.181.46|:443...
HTTP request sent, awaiting response... 200 OK
Length: 237168319 (226M) [application/octet-stream]
Saving to: 'ccle_vs_gdsc.tgz'

ccle_vs_gdsc.tgz  100%[=====>] 226.18M  11.4MB/s   in 22s

2025-03-14 14:13:16 (10.2 MB/s) - 'ccle_vs_gdsc.tgz' saved [237168319/2371
68319]

ccle_vs_gdsc/
ccle_vs_gdsc/train/
ccle_vs_gdsc/train/clin.csv
ccle_vs_gdsc/train/mutation.csv
ccle_vs_gdsc/train/cnv.csv
ccle_vs_gdsc/train/rna.csv
ccle_vs_gdsc/test/
ccle_vs_gdsc/test/clin.csv
ccle_vs_gdsc/test/mutation.csv
ccle_vs_gdsc/test/cnv.csv
ccle_vs_gdsc/test/rna.csv
```

```
In [22]: # Import necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import os
import glob
from sklearn.manifold import TSNE
import json
```

Use flexynesis on the command-line to predict drug responses for "Erlotinib". Write a bash script to run the following experiments: Try a combination of:

- a) different architectures: e.g. DirectPred, Supervised VAE, GNN (Test at least 2 of these).
- b) data type combinations (e.g. mutation, mutation + rna, mutation + cnv) (Test at least 2 of these)
- c) fusion methods: early, intermediate (applies only to tools other than GNN)

```
#!/bin/bash
```

```
DATA_PATH="./ccle_vs_gdsc"
DRUG="Erlotinib"
OUTPUT_DIR="./results"
FEATURE_PERCENTAGE=10 # Using 10% of features as suggested

mkdir -p $OUTPUT_DIR

MODEL_CLASSES=("DirectPred" "supervised_vae" "GNN")
DATA_TYPES=("mutation" "mutation,rna" "mutation,cnv")
FUSION_TYPES=("early" "intermediate")
GNN_CONV_TYPES=("GC" "SAGE")

for MODEL in "${MODEL_CLASSES[@]:0:2}"; do
  for DATA in "${DATA_TYPES[@]}"; do
    for FUSION in "${FUSION_TYPES[@]}"; do
      # Define experiment name
      EXP_NAME="${MODEL}_${DATA//,/}__${FUSION}"
      EXP_OUTDIR="${OUTPUT_DIR}/${EXP_NAME}"

      echo "Running experiment: $EXP_NAME"

      # Create output directory if it doesn't exist
      mkdir -p "$EXP_OUTDIR"

      # Run flexynesis with the current configuration
      flexynesis \
        --data_path $DATA_PATH \
        --model_class $MODEL \
        --data_types $DATA \
        --fusion_type $FUSION \
        --target_variables $DRUG \
        --features_top_percentile $FEATURE_PERCENTAGE \
        --variance_threshold $FEATURE_PERCENTAGE \
        --hpo_iter 15 \
        --outdir "$EXP_OUTDIR" \
        --prefix "$EXP_NAME" \
        --use_loss_weighting True
    done
  done
done

for DATA in "${DATA_TYPES[@]}"; do
  for CONV_TYPE in "${GNN_CONV_TYPES[@]}"; do
    # Define experiment name
```

```

EXP_NAME="GNN_${DATA//,/}_${CONV_TYPE}"
EXP_OUTDIR="${OUTPUT_DIR}/${EXP_NAME}"

echo "Running experiment: $EXP_NAME"

# Create output directory if it doesn't exist
mkdir -p "$EXP_OUTDIR"

# Run flexynesis with GNN configuration
flexynesis \
  --data_path $DATA_PATH \
  --model_class GNN \
  --gnn_conv_type $CONV_TYPE \
  --data_types $DATA \
  --fusion_type early \
  --target_variables $DRUG \
  --features_top_percentile $FEATURE_PERCENTAGE \
  --variance_threshold $FEATURE_PERCENTAGE \
  --hpo_iter 15 \
  --outdir "$EXP_OUTDIR" \
  --prefix "$EXP_NAME" \
  --use_loss_weighting True

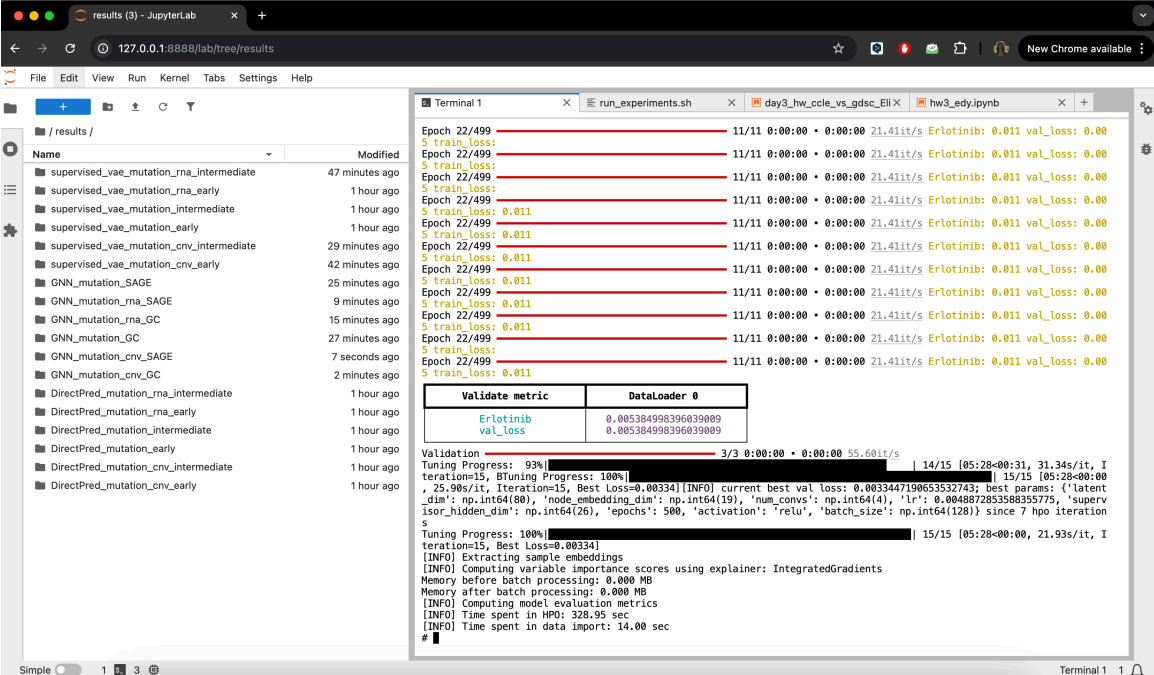
done
done

ls ./results/

```

In [69]: `from IPython.display import Image`
`Image(filename="./step1.png")`

Out [69]:



The screenshot shows a JupyterLab interface. On the left, a file browser displays a list of experiment results under the path `/results/`. The results are organized into folders for different drugs and models, such as `supervised_vae_mutation_rna_intermediate`, `supervised_vae_mutation_rna_early`, `supervised_vae_mutation_intermediate`, `supervised_vae_mutation_early`, `supervised_vae_mutation_cnv_intermediate`, `supervised_vae_mutation_cnv_early`, `GNN_mutation_SAGE`, `GNN_mutation_rna_SAGE`, `GNN_mutation_rna_GC`, `GNN_mutation_GC`, `GNN_mutation_cnv_SAGE`, `GNN_mutation_cnv_GC`, `DirectPred_mutation_rna_intermediate`, `DirectPred_mutation_rna_early`, `DirectPred_mutation_intermediate`, `DirectPred_mutation_early`, `DirectPred_mutation_cnv_intermediate`, and `DirectPred_mutation_cnv_early`. On the right, a terminal window displays the output of the `flexynesis` command. The logs show training progress for Erlotinib, including training loss, validation loss, and various metrics over 15 iterations. The logs also show the best parameters found during the search.

Import the results of the experiments from step 3, and rank the experiments based on performance (pearson_corr) Which combination yields the best results?

In [14]: `import os`
`import glob`

```
import pandas as pd

# Main path :
results_dir = "./results"
all_results = []

for folder in os.listdir(results_dir):
    folder_path = os.path.join(results_dir, folder)

    if not os.path.isdir(folder_path):
        continue

    csv_files = glob.glob(os.path.join(folder_path, "*stat*.csv"))

    for csv_file in csv_files:
        df = pd.read_csv(csv_file)
        pearson_row = df[df["metric"] == "pearson_corr"]

        if not pearson_row.empty:
            pearson_value = pearson_row["value"].values[0]
            all_results.append({"folder": folder, "pearson_corr": pearson_value})
```

```
In [15]: results_df = pd.DataFrame(all_results)
results_df = results_df.sort_values(by="pearson_corr", ascending=False)
```

```
In [17]: results_df
```

Out [17]:

	folder	pearson_corr
13	DirectPred_mutation_rna_early	0.440513
14	DirectPred_mutation_rna_intermediate	0.429166
0	supervised_vae_mutation_rna_intermediate	0.417457
8	supervised_vae_mutation_rna_early	0.415265
1	GNN_mutation_rna_GC	0.370119
12	GNN_mutation_rna_SAGE	0.324514
16	supervised_vae_mutation_cnv_early	0.198822
10	GNN_mutation_cnv_GC	0.198311
11	DirectPred_mutation_cnv_intermediate	0.161136
7	supervised_vae_mutation_cnv_intermediate	0.157989
2	DirectPred_mutation_cnv_early	0.121997
3	GNN_mutation_GC	0.082084
17	DirectPred_mutation_intermediate	0.074486
9	GNN_mutation_cnv_SAGE	0.052161
4	supervised_vae_mutation_early	0.031970
5	GNN_mutation_SAGE	0.023756
15	DirectPred_mutation_early	0.013787
6	supervised_vae_mutation_intermediate	0.013539

In [16]: `best_model = results_df.iloc[0]`

```
print("Best Model:")
print(best_model)
```

```
Best Model:
folder      DirectPred_mutation_rna_early
pearson_corr      0.440513
Name: 13, dtype: object
```

In [28]: `flexynesis.print_summary_stats(train_dataset)`

Summary for variable: Crizotinib
Numerical Variable Summary: Median = 0.0352093531894524, Mean = 0.054255634587440156

Summary for variable: Dovitinib
Numerical Variable Summary: Median = 0.07464228066682425, Mean = 0.08821146252683426

Summary for variable: Erlotinib
Numerical Variable Summary: Median = 0.0403443205178365, Mean = 0.06692524772422281

Summary for variable: Irinotecan
Numerical Variable Summary: Median = 0.3920620379485415, Mean = 0.39017780935837576

Summary for variable: L-685458
Numerical Variable Summary: Median = 0.0138021275751136, Mean = 0.030322647554898396

Summary for variable: Lapatinib
Numerical Variable Summary: Median = 0.0394095700756828, Mean = 0.0675491054885661

Summary for variable: LBW242
Numerical Variable Summary: Median = 0.0547794573417837, Mean = 0.07855441159620391

Summary for variable: Nilotinib
Numerical Variable Summary: Median = 0.03057075166965985, Mean = 0.05153153782549488

Summary for variable: Nutlin-3
Numerical Variable Summary: Median = 0.0286908589018313, Mean = 0.03810224795989919

Summary for variable: Nvp-aew541
Numerical Variable Summary: Median = 0.08490990819417804, Mean = 0.10457684338476954

Summary for variable: Nvp-tae 684
Numerical Variable Summary: Median = 0.11362412539153, Mean = 0.13236452160732368

Summary for variable: Paclitaxel
Numerical Variable Summary: Median = 0.7018025588210894, Mean = 0.6841379341400685

Summary for variable: Palbociclib
Numerical Variable Summary: Median = 0.051872041608588146, Mean = 0.06825555015423597

Summary for variable: Panobinostat
Numerical Variable Summary: Median = 0.579730990993632, Mean = 0.5882682741471223

Summary for variable: PD-0325901
Numerical Variable Summary: Median = 0.21022026577672698, Mean = 0.2554964357077726

Summary for variable: Pha-665752
Numerical Variable Summary: Median = 0.0285848706863142, Mean = 0.04638751619785537

Summary for variable: PLX4720
Numerical Variable Summary: Median = 0.0178014667778967, Mean = 0.044434097002627286

Summary for variable: Raf265(chir-265)
Numerical Variable Summary: Median = 0.171686832440114, Mean = 0.17934439587170292

Summary for variable: Saracatinib
Numerical Variable Summary: Median = 0.0815107683318432, Mean = 0.09873959568146576

Summary for variable: Selumetinib
Numerical Variable Summary: Median = 0.1239289409762335, Mean = 0.16281416102974458

Summary for variable: Sorafenib
Numerical Variable Summary: Median = 0.01503532722932145, Mean = 0.027591934840777415

Summary for variable: Tanespimycin
Numerical Variable Summary: Median = 0.4203236423453, Mean = 0.4213244657773919

Summary for variable: Topotecan
Numerical Variable Summary: Median = 0.353478480952629, Mean = 0.3546567702335896

Summary for variable: Vandetanib
Numerical Variable Summary: Median = 0.0992853166302971, Mean = 0.11649212727237457

Summary for variable: tissueid
Categorical Variable Summary:
Label: Ampulla of Vater, Count: 2
Label: Biliary Tract, Count: 6
Label: Bladder/Urinary Tract, Count: 28
Label: Bone, Count: 27
Label: Bowel, Count: 62
Label: Breast, Count: 59
Label: CNS/Brain, Count: 69
Label: Cervix, Count: 1
Label: Esophagus/Stomach, Count: 65
Label: Head and Neck, Count: 34
Label: Kidney, Count: 35
Label: Liver, Count: 26
Label: Lung, Count: 182
Label: Lymphoid, Count: 126
Label: Myeloid, Count: 52
Label: Other, Count: 5
Label: Ovary/Fallopian Tube, Count: 52
Label: Pancreas, Count: 45
Label: Peripheral Nervous System, Count: 17
Label: Pleura, Count: 11
Label: Prostate, Count: 8
Label: Skin, Count: 61

Label: Soft Tissue, Count: 16

Label: Thyroid, Count: 13

Label: Uterus, Count: 29

2. Explore the embeddings from the best model:

```
In [31]: import matplotlib.pyplot as plt
         from sklearn.decomposition import PCA
         from sklearn.manifold import TSNE
```

```
In [ ]: # ./results/DirectPred_mutation_rna_early/
```

```
In [36]: best_model
```

```
Out[36]: folder          DirectPred_mutation_rna_early
         pearson_corr          0.440513
         Name: 13, dtype: object
```

```
In [37]: best_model_pearcorr = results_df.iloc[13]['pearson_corr']
```

```
In [38]: best_model_pearcorr
```

```
Out[38]: np.float64(0.0521614370460921)
```

```
In [44]: train_embeddings_path = os.path.join('./results/DirectPred_mutation_rna_e
         test_embeddings_path = os.path.join('./results/DirectPred_mutation_rna_ea
```

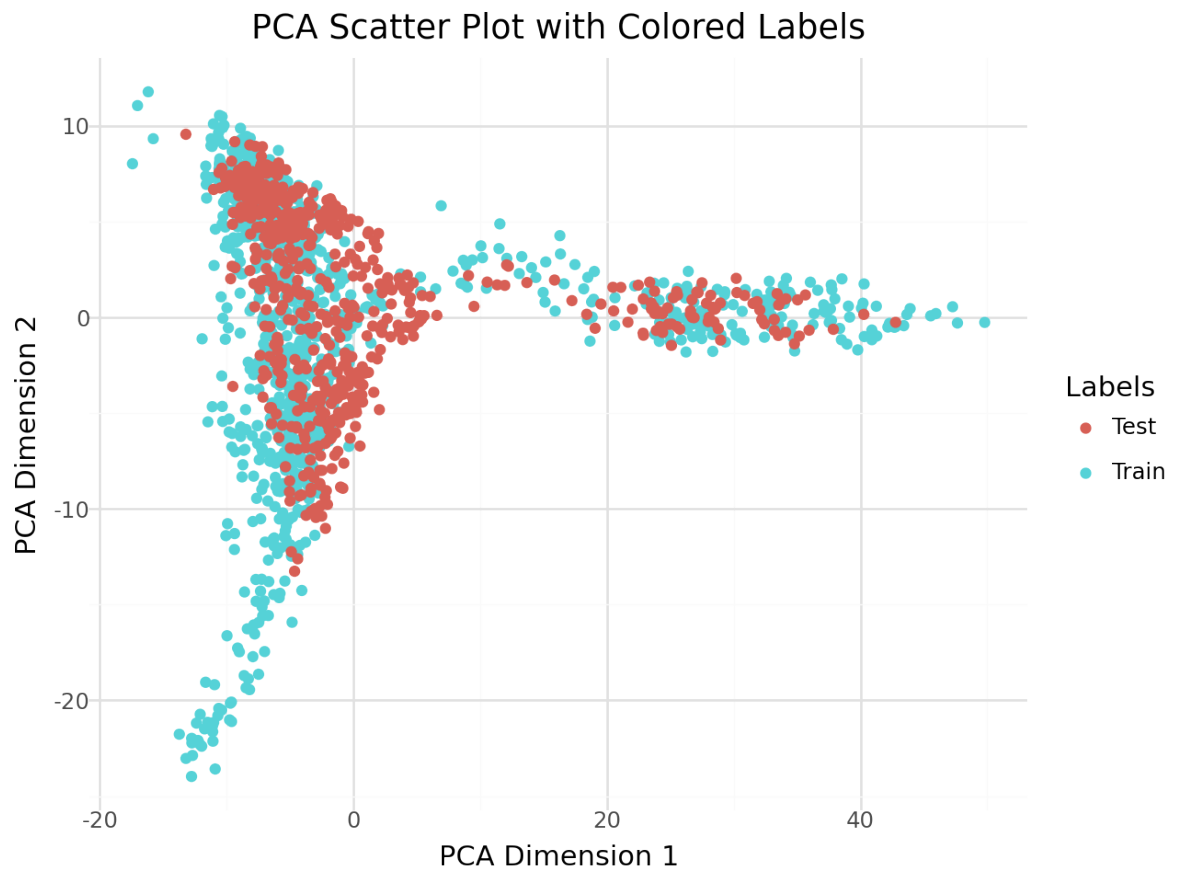
```
In [45]: # Load the embeddings
         train_embeddings = pd.read_csv(train_embeddings_path)
         test_embeddings = pd.read_csv(test_embeddings_path)
```

```
In [47]: train_embeddings_numeric = train_embeddings.select_dtypes(include=['float
         test_embeddings_numeric = test_embeddings.select_dtypes(include=['float64
```

```
In [48]: # Perform PCA or t-SNE to visualize embeddings (dimensionality reduction)
         pca = PCA(n_components=2)
         train_pca = pca.fit_transform(train_embeddings_numeric)
         test_pca = pca.transform(test_embeddings_numeric)
```

```
In [50]: from flexynesis import plot_dim_reduced
```

```
In [51]: all_pca = np.vstack([train_pca, test_pca])
         labels = ['Train'] * len(train_pca) + ['Test'] * len(test_pca)
         plot_dim_reduced(all_pca, labels=labels, method='pca', color_type='catego
```

3. Analyze feature importance from the best model:

```
In [59]: feature_importance = pd.read_csv('./results/DirectPred_mutation_rna_early')  
In [60]: print(feature_importance)
```

	target_variable	target_class	target_class_label	layer	name \
0	Erlotinib	0	NaN	all	SOX10
1	Erlotinib	0	NaN	all	TNFRSF17
2	Erlotinib	0	NaN	all	IKZF1
3	Erlotinib	0	NaN	all	BTK
4	Erlotinib	0	NaN	all	VPREB3
...	...	...	...	...	...
1262	Erlotinib	0	NaN	all	CDKN2A
1263	Erlotinib	0	NaN	all	PIK3R1
1264	Erlotinib	0	NaN	all	TSC1
1265	Erlotinib	0	NaN	all	GNAS
1266	Erlotinib	0	NaN	all	ERBB2

	importance	explainer
0	0.000150	IntegratedGradients
1	0.000090	IntegratedGradients
2	0.000085	IntegratedGradients
3	0.000481	IntegratedGradients
4	0.000154	IntegratedGradients
...	...	...
1262	0.000119	IntegratedGradients
1263	0.000235	IntegratedGradients
1264	0.000393	IntegratedGradients
1265	0.000807	IntegratedGradients
1266	0.000092	IntegratedGradients

[1267 rows x 7 columns]

```
In [61]: top_10_features = feature_importance.sort_values(by='importance', ascending=False)
```

```
In [62]: print("Top 10 important features:")
print(top_10_features)
```

Top 10 important features:

	target_variable	target_class	target_class_label	layer	name \
442	Erlotinib	0	NaN	all	ICA1
1258	Erlotinib	0	NaN	all	KRAS
824	Erlotinib	0	NaN	all	HKDC1
1106	Erlotinib	0	NaN	all	XDH
121	Erlotinib	0	NaN	all	EGFR
586	Erlotinib	0	NaN	all	CTSV
623	Erlotinib	0	NaN	all	ARL14
474	Erlotinib	0	NaN	all	ACSL5
924	Erlotinib	0	NaN	all	STAP2
280	Erlotinib	0	NaN	all	BICC1

	importance	explainer
442	0.002206	IntegratedGradients
1258	0.002204	IntegratedGradients
824	0.002177	IntegratedGradients
1106	0.002158	IntegratedGradients
121	0.002112	IntegratedGradients
586	0.001909	IntegratedGradients
623	0.001885	IntegratedGradients
474	0.001855	IntegratedGradients
924	0.001846	IntegratedGradients
280	0.001843	IntegratedGradients

Literature

ICA1, KRAS, HKDC1, XDH, EGFR, CTSV, ARL14, ACSL5, STAP2 ve BICC1

Especially; KRAS, EGFR, CTSV, ACSL5, and XDH are significant cancer-related genes, and their mutations contribute to the progression of multiple cancer types, particularly **lung**, colorectal, and pancreatic cancer.

- **Lung Cancer**

- EGFR - KRAS ->> [Zhu et al, 2008](#)
- EGFR - KRAS ->> [D'Angelo et al, 2012](#)
- EGFR - KRAS ->> [Yoshizawa et al, 2013](#)
- EGFR - CTSV ->> [Mataki et al, 2014](#)